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DSCI 550

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Homework 3

1. a. Overfitting is when a model is too sensitive to specific training data and captures random noise in the data rather than actual patterns. This causes it to fit training data much more accurately than test data.
- b. Some ways to avoid overfitting are: collecting more data, Cross-validation, and regularization.
- c. Common reasons ML fails include underfitting, bad data and data leakage.

2. a. Cross-Validation in ML is a process where you train and assess your data on consecutive folds of it (subsets). This reduces overfitting and provides a more custom and accurate way to assess models.

b. False. Classifier A's 90% train but 70% test suggests overfitting. Classifier B shows no such signs and outperforms A on the test set (78-70).

Bayes: $P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$

q. Find $P(\text{patient has cancer} | \text{positive test})$

$$= \frac{P(\text{positive test} | \text{cancer}) \cdot P(\text{cancer})}{P(\text{positive test})}$$

$$= \frac{0.98 \cdot 0.008}{(.98 \cdot 0.008 + .03 \cdot .992)}$$

$$= \frac{.00784}{.0376}$$

$$\approx 0.2085$$

Since this patient had a positive test, there is about a 20% chance they have cancer.

b. find $P(\text{no cancer} | \text{positive test})$:

$$= \frac{P(\text{positive test} | \text{no cancer}) \cdot P(\text{no cancer})}{P(\text{positive test})}$$

$$= \frac{.03 \cdot .992}{(.98 \cdot .008 + .03 \cdot .992)}$$

$$= \frac{.0297}{.0376}$$

$$\approx 0.7915$$

There is about an 80% chance this patient does not have cancer given the positive test

C. The diagnosis is no cancer
but get tested again to be
sure.

4. High fever, no sneezing, runny nose

$$P(\text{Flu}) = 0.5$$

$$P(\text{Sneeze} = \text{No} | \text{Flu}) = 0.3$$

$$P(\text{Allergy}) = 0.3$$

$$P(\text{Sneeze} = \text{No} | \text{Allergy}) = 0.5$$

$$P(\text{Other}) = 0.2$$

$$P(\text{Sneeze} = \text{No} | \text{Other}) = 0.25$$

$$P(\text{Fever} = \text{Yes} | \text{Flu}) = 0.8 \quad P(\text{RN} | \text{Flu}) = 0.9$$

$$\rightarrow P(\text{Fever} = \text{Yes} | \text{Allergy}) = 0 \quad P(\text{RN} | \text{Allergy}) = 0$$

$$P(\text{Fever} = \text{Yes} | \text{Other}) = 0.5 \quad P(\text{RN} | \text{Other}) = 0.25$$

$$\boxed{\text{Flu}}: 0.5 (0.8 \cdot 0.3 \cdot 0.9) \approx 0.108$$

$$\text{Allergy}: 0.3 (0 \cdot 0.5 \cdot 0) = 0$$

$$\text{Other}: 0.2 (0.25 \cdot 0.5 \cdot 0.25) \approx 0.006$$

Using Naive Bayes, we conclude our patient likely has the flu.

5.

X = num shopping centers
 Y = retail sales in \$bn
a. $n = 8$

$$\sum X = 7490 \quad \sum Y = 187.3$$

$$\sum XY = 332620.8 \quad \sum X^2 = 12207356$$

$$\sum Y^2 = 9231.17$$

$$\text{slope} = \frac{n \cdot \sum XY - \sum X \cdot \sum Y}{n \cdot \sum X^2 - (\sum X)^2}$$

$$= \frac{8 \cdot 332620.8 - 7490 \cdot 187.3}{8 \cdot 12207356 - (7490)^2}$$

$$= \frac{2660966.4 - 1402877}{97658848 - 56100100}$$

$$= \frac{1258089.4}{41558748}$$

$$\approx 0.0303$$

$$\text{Intercept: } b = \frac{\sum Y - m \cdot \sum X}{n}$$

$$b = \frac{187.3 - (0.03027 \cdot 7490)}{8}$$

$$= \frac{187.3 - 226.7223}{8}$$

$$= \frac{-39.4223}{8}$$

$$= -4.93$$

Equation

$$\hat{Y} = 0.0303X - 4.93$$

b. Slope: For every additional shopping center in a state, the retail sales of that state will increase by roughly

\$30 million,

Intercept: A state with no shopping centers will have roughly ~~\$~~\$4.93 trillion in retail sales (this is illogical).

$$C. \quad r = \frac{n \cdot \sum XY - \sum X \cdot \sum Y}{\sqrt{(n \cdot \sum(X^2) - (\sum X)^2) \cdot (n \cdot \sum(Y^2) - (\sum Y)^2)}}$$

$$r = \frac{8 \cdot 332\,620.8 - 7440 \cdot 187.3}{\sqrt{(8 \cdot 12\,207\,356 - (7440)^2) \cdot (8 \cdot 9231.7 - (187.3)^2)}}$$

$$r = \frac{2\,660\,466.4 - 1\,402\,877}{\sqrt{(97\,658\,848 - 56\,100\,100) \cdot (73\,849.36 - 35\,081)}}$$

$$r = \frac{1\,258\,089.4}{\sqrt{41\,558\,748 \cdot 38\,768.07}}$$

$$= \frac{1\,258\,089.4}{1\,269\,311.83} \quad \text{Interpret} \downarrow$$

$$= 0.9912$$

$$\boxed{r^2 \approx 0.982}$$

Approximately 98.2% of the variance in retail sales of a state can be explained by the number of shopping centers in the state.

$$d. \text{ Error} = Y - \hat{Y}$$

$$Y \text{ for } 1000: 23.0$$

$$\begin{aligned}\hat{Y} \text{ for } 1200 &= 0.0303(1200) - 4.93 \\ &= 31.43\end{aligned}$$

$$\text{Error} = 23 - 31.43 = \text{\$}8.43 \text{ bn}$$

e. You cannot accurately predict $X=100$ since our X range in the training set is ≥ 370 . This would be extrapolation.

6.

(1) Frequent Items: (90%.)

A: 4/5
B: 3/5
C: 5/5
D: 4/5
E: 2/5

} all frequent so must
look at all 2 item sets

2 item sets

✓ AB: 2/5 ✗ BE: 1/5
✓ AC: 4/5 ✓ CD: 4/5
✓ AD: 3/5 ✓ CE: 2/5
✗ AE: 1/5 ✓ DE: 2/5
✓ BC: 3/5
✓ BD: 2/5

Frequent 2 item
sets are:

AB, AC, AD,

BC, BD, CD,

CE, DE. We don't
need to check 3 item
sets w/ AE or BE

3 item sets (not including AE/BE)

↓ ABC: 2/5
× ABD: 1/5
↓ ACD: 3/5
↓ BCD: 2/5
↓ CDE: 2/5

Frequent 3 item sets include:

ABC, ACD, BCD, CDE. We only need to check 4 item sets with these combos.

(not checking ABCE, ACE, ADE, BCE, BDE)

4 item sets

ABCD - not checked (ABD)

ABCE - not checked (BCE)

ABDE - not checked (BDE)

ACDE - not checked (ACE)

BCE - not checked (BCE)

So there are no frequent 4 item sets.

6(2)

$$1. \frac{\text{support}\{B, C, E\}}{\text{support}\{B, C\}} = \frac{30\%}{40\%} = 75\%.$$

75 > 70 so this rule is strong.

$$2. \frac{\text{support}\{B, C, E\}}{\text{support}\{B\}} = \frac{30\%}{50\%} = 60\%.$$

60 < 70 so this rule is not strong

7.

Single Link

$\{3, 4\}$ smallest gap .4

$$d(\{3, 4\}, 1) = 1.12 \leftarrow$$

$$d(\{3, 4\}, 2) = 2.12$$

$$d(\{3, 4\}, 5) = 1.28$$

$\{1, 3, 4\}$ is next cluster

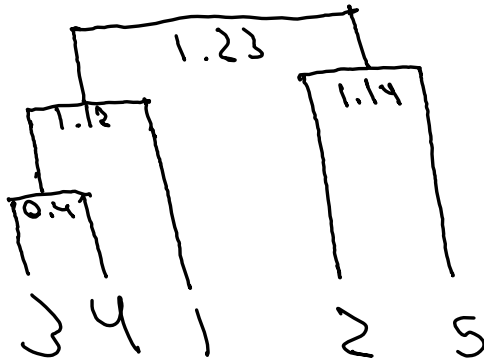
2 and 5 closer to each other than $\{1, 3, 4\}$

$\{2, 5\}$ is next cluster

Final merge of $\{1, 3, 4\}$ and $\{2, 5\}$

has $d(1.23)$

Dendrogram



Complete Link

Same start with 3, 4 being closest

$$d(1, \{3, 4\}) = 1.34$$

$$d(2, \{3, 4\}) = 2.16 \text{ , all bigger than}$$

$$d(5, \{3, 4\}) = 1.42 \quad 1.14 \text{ between } 2, 5$$

$\{2, 5\}$ is second cluster

$$d(1, \{2, 5\}) = 1.24$$

so next cluster is

$$\{1, 2, 5\} \text{ since } 1.24 < 1.34,$$

then we merge all at 2.16

$$(d(2, 3))$$

Dendrogram

